

T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after F

This stage has active field content $\text{SM}+S_1+S_2$. The report combines the renormalisable RGE running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out F . Throughout, $16\pi^2\mu dX/d\mu = \beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Renormalisable EFT1 couplings

$$\beta_{g_Y}^{(1)}$$

$$16\pi^2\mu\frac{dg_Y}{d\mu} = \beta_{g_Y}^{(1)}$$

Column Representative term structure	
T1	$\frac{43g_Y^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{43g_Y^3}{6}$

$$\beta_{g_2}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{17g_2^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17g_2^3}{6}$

$$\beta_{g_3}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column	Representative term structure	
T1	$-7g_3^3$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-7g_3^3$

$\beta_{Y_u}^{(1)}$

$16\pi^2\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$-3\text{Tr}\Big(Y_d\left(Y_d\right)^{\dagger}\Big)\left(Y_u\right)_{ij}$
T2	$+ \text{Tr}\Big(Y_e\left(Y_e\right)^{\dagger}\Big)\left(Y_u\right)_{ij}$
T3	$+3\text{Tr}\Big(Y_u\left(Y_u\right)^{\dagger}\Big)\left(Y_u\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\text{Tr}\Big(Y_d\left(Y_d\right)^{\dagger}\Big)\left(Y_u\right)_{ij}$	$+ \text{Tr}\Big(Y_e\left(Y_e\right)^{\dagger}\Big)\left(Y_u\right)_{ij}$	$+3\text{Tr}\Big(Y_u\left(Y_u\right)^{\dagger}\Big)\left(Y_u\right)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$
T3	$-8g_3^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}gY^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17}{12}gY^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

$16\pi^2\,\mu\,\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}gY^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{5}{12}gY^2(Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

$16\pi^2\,\mu\,\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$
T3	$+\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$
T2	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$
T3	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$

$\beta_{m_{S_1}^2}^{(1)}$

$16\pi^2\mu\frac{dm_{S_1}^2}{d\mu}=\beta_{m_{S_1}^2}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$-\frac{9g_2^2m_{S_1}^2}{2}$
T2	$-\frac{3g_Y^2m_{S_1}^2}{2}$
T3	$+4\lambda_{12}m_{S_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2m_{S_1}^2}{2}$	$-\frac{3g_Y^2m_{S_1}^2}{2}$	$+4\lambda_{12}m_{S_2}^2$

Terms 2 of 2

Column	Representative term structure
T1	$+6\lambda_{S_1} m_{S_1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{S_1} m_{S_1}^2$

$\beta_{m_{S_2}^2}^{(1)}$

$16\pi^2\mu\frac{dm_{S_2}^2}{d\mu}=\beta_{m_{S_2}^2}^{(1)}$

Terms 1 of 2

Column Representative term structure	
T1	$-\frac{9g_2^2m_{S_2}^2}{2}$
T2	$-\frac{3g_Y^2m_{S_2}^2}{2}$
T3	$+4\lambda_{12}m_{S_1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2m_{S_2}^2}{2}$	$-\frac{3g_Y^2m_{S_2}^2}{2}$	$+4\lambda_{12}m_{S_1}^2$

Terms 2 of 2

Column	Representative term structure
T1	$+6\lambda_{S_2} m_{S_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{S_2} m_{S_2}^2$

$\beta_{\lambda_H}^{(1)}$

$16\pi^2\mu\frac{d\lambda_H}{d\mu}=\beta_{\lambda_H}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$
T3	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$

Terms 2 of 6

Column	Representative term structure
T1	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$
T2	$+8\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$
T3	$+4\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+8\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$	$+4\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$

Terms 3 of 6

Column	Representative term structure
T1	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$
T2	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$

Terms 4 of 6

Column	Representative term structure
T1	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$
T2	$-9g_2^2\lambda_H$
T3	$+\frac{3g_Y^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$

Terms 5 of 6

Column	Representative term structure
T1	$+12\lambda_H^2$
T2	$+4\lambda_{H1}^2$
T3	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$

Terms 6 of 6

Column	Representative term structure
T1	$+4\lambda_{H2}^2$
T2	$+\frac{\lambda_{H2}^{Adj^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H2}^2$	$+\frac{\lambda_{H2}^{Adj^2}}{4}$

$$\beta_{\lambda_{S_1}}^{(1)}$$

$$_{16\pi^2}\mu\frac{d\lambda_{S_1}}{d\mu}=\beta_{\lambda_{S_1}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$
T2	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$
T3	$+24\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\left(\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$	$+24\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\left(\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\right)^*$

Terms 2 of 4

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$
T3	$-9g_2^2 \lambda_{S_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{9g_2^4}{4}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$	$-9g_2^2 \lambda_{S_1}$

Terms 3 of 4

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{4}$
T2	$+4\lambda_{12}^2$
T3	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{3g_Y^4}{4}$	$+4\lambda_{12}^2$	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_{H1}^2$
T2	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$
T3	$+12\lambda_{S1}^2$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H1}^2$	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$	$+12\lambda_{S1}^2$

$\beta_{\lambda_{S_2}}^{(1)}$

$_{16\pi^2}\mu\frac{d\lambda_{S_2}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$8\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$
T2	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$
T3	$+24\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$8\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$	$+24\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*$

Terms 2 of 4

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$
T3	$-9g_2^2 \lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$	$-9g_2^2 \lambda_{S_2}$

Terms 3 of 4

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{4}$
T2	$+4\lambda_{12}^2$
T3	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{3g_Y^4}{4}$	$+4\lambda_{12}^2$	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_{H2}^2$
T2	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3	$+12\lambda_{S2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H2}^2$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	$+12\lambda_{S2}^2$

$\beta_{\lambda_{H1}}^{(1)}$

$16\pi^2\,\mu\,\frac{d\lambda_{H1}}{d\mu}=\beta_{\lambda_{H1}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$
T3	$+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{H H^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{H H^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Terms 2 of 5

Column	Representative term structure
T1	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^*)$
T2	$+ 6 \lambda_{T3} (\lambda_{T3})^*$
T3	$+ \frac{9 g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^*) + 6 \lambda_{T3} (\lambda_{T3})^*$		$+ \frac{9 g_2^4}{4}$

Terms 3 of 5

Column	Representative term structure
T1	$-9g_2^2\lambda_{H1}$
T2	$+\frac{3g_Y^4}{4}$
T3	$-3g_Y^2\lambda_{H1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda_{12}\lambda_{H2}$
T2	$+6\lambda_H\lambda_{H1}$
T3	$+4\lambda_{H1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$

Terms 5 of 5

Column	Representative term structure
T1	$+6\lambda_{H1}\lambda_{S1}$
T2	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{H1}\lambda_{S1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$

$\beta_{\lambda_{H2}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{H2}}{d\mu}=\beta_{\lambda_{H2}}^{(1)}$

Terms 1 of 5

Column	Representative term structure	
T1	$\lambda_{H2}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	
T2	$+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	
T3	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H2}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*)$

Terms 2 of 5

Column	Representative term structure
T1	$+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$
T2	$+6\lambda_{T3} \left(\lambda_{T3}\right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$	$+6\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$

Terms 3 of 5

Column	Representative term structure
T1	$-9g_2^2\lambda_{H2}$
T2	$+\frac{3g_Y^4}{4}$
T3	$-3g_Y^2\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H2}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H2}$

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda_{12}\lambda_{H1}$
T2	$+6\lambda_H\lambda_{H2}$
T3	$+4\lambda_{H2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$

Terms 5 of 5

Column	Representative term structure
T1	$+6\lambda_{H2}\lambda_{S2}$
T2	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{H2}\lambda_{S2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$

$\beta_{\lambda_{12}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$
T3	$+9\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$	$+9\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$

Terms 2 of 5

Column	Representative term structure
T1	$+9\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right)$
T2	$+3\lambda_{T3} \left(\lambda_{T3} \right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+9\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) +3\lambda_{T3} \left(\lambda_{T3} \right)^*$		$+\frac{9g_2^4}{4}$

Terms 3 of 5

Column	Representative term structure
T1	$-9g_2^2\lambda_{12}$
T2	$+\frac{3g_Y^4}{4}$
T3	$-3g_Y^2\lambda_{12}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{12}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{12}$

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda_{12}^2$
T2	$+6\lambda_{12}\lambda_{S_1}$
T3	$+6\lambda_{12}\lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}^2$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$

Terms 5 of 5

Column	Representative term structure
T1	$+ \frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T2	$+4\lambda_{H1}\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{3\lambda_{12}^{\text{Adj}^2}}{4}$	$+4\lambda_{H1}\lambda_{H2}$

$$\beta_{\lambda_{T3}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{d\lambda_{T3}}{d\mu}=\beta_{\lambda_{T3}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure	
T1	λ_{T3}	$\left(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$
T2	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$
T3	$+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}$	$\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{T3}\left(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$

Terms 2 of 4

Column	Representative term structure
T1	$+4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{H H^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$
T2	$-9g_2^2\lambda_{T3}$
T3	$-3g_Y^2\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{H H^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)-9g_2^2\lambda_{T3}$		$-3g_Y^2\lambda_{T3}$

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_{12}\lambda_{T3}$
T2	$+2\lambda_H\lambda_{T3}$
T3	$+4\lambda_{H1}\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_{H2}\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H2}\lambda_{T3}$

$$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{64}{3}\lambda_{S_1^\dagger S_2^\dagger}^2\left(\lambda_{S_1^\dagger S_2^\dagger}^2\right)^*$
T2	$+\frac{4}{3}\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\left(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\right)^*+\left(\lambda_{S_1 S_1^\dagger S_2^\dagger}^2\right)^*\right)$
T3	$+\frac{4}{3}\lambda_{S_1 S_1^\dagger S_2^\dagger}^2\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\right)^*+5\left(\lambda_{S_1 S_1^\dagger S_2^\dagger}^2\right)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{64}{3}\lambda_{S_1^\dagger S_2^\dagger}^2\left(\lambda_{S_1^\dagger S_2^\dagger}^2\right)^*$	$+\frac{4}{3}\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\left(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\right)^*+\left(\lambda_{S_1 S_1^\dagger S_2^\dagger}^2\right)^*\right)+\frac{4}{3}\lambda_{S_1 S_1^\dagger S_2^\dagger}^2\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\right)^*+5\left(\lambda_{S_1 S_1^\dagger S_2^\dagger}^2\right)^*\right)$	

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$
T2	$+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$
T3	$-9g_2^2\lambda_{12}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$	$-9g_2^2\lambda_{12}^{\text{Adj}}$

Terms 3 of 4

Column	Representative term structure
T1	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T2	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$
T3	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

$$\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

$$_{16\pi^2}\mu\frac{d\lambda_{H1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H1}^{\text{Adj}}(6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$
T2	$+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$
T3	$-\frac{4}{3}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H1}^{\text{Adj}}(6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$-\frac{4}{3}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$
T2	$+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$
T3	$+g_Y^2\left(-6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(-6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$	

Terms 3 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_{H1}^{\text{Adj}}$
T2	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$

Terms 4 of 4

Column	Representative term structure
T1	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$
T2	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$

$$\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$$

$$_{16\pi^2}\mu^{-2}\frac{d\lambda_{H2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$
T2	$-4\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-4\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*)$

Terms 2 of 4

Column	Representative term structure
T1	$+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$
T2	$+16\lambda_{T3} \left(\lambda_{T3}\right)^*$
T3	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H2}^{\text{Adj}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$	$+16\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H2}^{\text{Adj}}\right)$

Terms 3 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_{H2}^{\text{Adj}}$
T2	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$

Terms 4 of 4

Column	Representative term structure
T1	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$
T2	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$

$$\beta_{\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}^{(1)}$$

$$16\pi^2\,\mu\frac{d\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} (6\, \text{Tr}\big(Y_d\, (Y_d)^\dagger\big) + 2\, \text{Tr}\big(Y_e\, (Y_e)^\dagger\big) + 6\, \text{Tr}\big(Y_u\, (Y_u)^\dagger\big))$
T2	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\, (\lambda_{T3})^*$
T3	$+2\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\, (\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} (6\, \text{Tr}\big(Y_d\, (Y_d)^\dagger\big) + 2\, \text{Tr}\big(Y_e\, (Y_e)^\dagger\big) + 6\, \text{Tr}\big(Y_u\, (Y_u)^\dagger\big))$	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\, (\lambda_{T3})^*$	$+2\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\, (\lambda_{T3})^*$

Terms 2 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T2	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$

Terms 3 of 4

Column	Representative term structure
T1	$+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T2	$+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$
T3	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S_1}$

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$

$$\beta_{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}^{(1)}}$$

$$16\pi^2\mu\frac{d\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{d\mu}=\beta_{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}^{(1)}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+6\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3	$-6\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*(\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+6\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-6\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*(\lambda_{T3})^*$

Terms 2 of 4

Column	Representative term structure	
T1	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	
T2	$-3g_Y^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	
T3	$+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-3g_Y^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Terms 3 of 4

Column	Representative term structure
T1	$+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$+\lambda_{H1}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger})$
T3	$+\lambda_{H1}^{\text{Adj}}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+\lambda_{H1}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger})$	$+\lambda_{H1}^{\text{Adj}}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$

Terms 4 of 4

Column	Representative term structure
T1	$+ \lambda_{H2} (4 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 3 \lambda_{S_1^\dagger S_2 S_2^\dagger 2})$
T2	$+ \lambda_{H2}^{\text{Adj}} \left(- \frac{3 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3 \lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{4} \right)$
T3	$- 6 \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_{H2} (4 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 3 \lambda_{S_1^\dagger S_2 S_2^\dagger 2})$	$+ \lambda_{H2}^{\text{Adj}} \left(- \frac{3 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3 \lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{4} \right)$	$- 6 \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{T3}$

$$\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$$

$$16\pi^2\,\mu\frac{d\lambda_{H^\dagger H^\dagger S_2 S_2}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{H^\dagger H^\dagger S_2 S_2} (6\, \text{Tr}\big(Y_d\, (Y_d)^\dagger\big) + 2\, \text{Tr}\big(Y_e\, (Y_e)^\dagger\big) + 6\, \text{Tr}\big(Y_u\, (Y_u)^\dagger\big))$
T2	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$
T3	$+2\, (\lambda_{T3})^*\, \big(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H^\dagger H^\dagger S_2 S_2} (6\, \text{Tr}\big(Y_d\, (Y_d)^\dagger\big) + 2\, \text{Tr}\big(Y_e\, (Y_e)^\dagger\big) + 6\, \text{Tr}\big(Y_u\, (Y_u)^\dagger\big))$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$	$+2\, (\lambda_{T3})^*\, \big(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\big)$

Terms 2 of 3

Column	Representative term structure
T1	$-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$
T2	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$

Terms 3 of 3

Column	Representative term structure
T1	$+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$
T2	$+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3	$+2\lambda_{H^\dagger H^\dagger S_2 S_2}\lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+2\lambda_{H^\dagger H^\dagger S_2 S_2}\lambda_{S_2}$

$$\beta_{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}^{(1)}$$

$$16\pi^2\,\mu\frac{d\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{d\mu}=\beta_{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$4\lambda_{S_1^\dagger^2 S_2^\dagger^2}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger^2}\right)^*+5\left(\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}\right)^*)$
T2	$-9g_2^2\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}$
T3	$-3g_Y^2\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_{S_1^\dagger^2 S_2^\dagger^2}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger^2}\right)^*+5\left(\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}\right)^*)$	$-9g_2^2\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}$	$-3g_Y^2\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}$

Terms 2 of 3

Column	Representative term structure
T1	$+ \lambda_{12} (6 \lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$
T2	$+ \lambda_{12}^{\text{Adj}} \left(\frac{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{5 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6} \right)$
T3	$+ 2 \lambda_{H1} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_{12} (6 \lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$	$+ \lambda_{12}^{\text{Adj}} \left(\frac{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{5 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6} \right)$	$+ 2 \lambda_{H1} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$
T2	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$
T3	$+12\lambda_{S_1}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$	$+12\lambda_{S_1}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}$

$$\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

$$16\pi^2\,\mu\frac{d\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}{d\mu}=\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$
T2	$-9g_2^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$
T3	$-3g_Y^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$	$-9g_2^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$	$-3g_Y^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$

Terms 2 of 4

Column	Representative term structure
T1	$+8\lambda_{12}\lambda_{S_1^\dagger 2 S_2^\dagger 2}$
T2	$+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^\dagger 2 S_2^\dagger 2}}{3}$
T3	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{12}\lambda_{S_1^\dagger 2 S_2^\dagger 2}$	$+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^\dagger 2 S_2^\dagger 2}}{3}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_{S_1} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$
T2	$+2\lambda_{S_1^\dagger 2 S_2^\dagger 2} \lambda_{S_2}$
T3	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{S_1} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$	$+2\lambda_{S_1^\dagger 2 S_2^\dagger 2} \lambda_{S_2}$	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}$
T2	$+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}$	$+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$

$$\beta_{\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}^{(1)}$$

$$16\pi^2\,\mu\frac{d\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}{d\mu}=\beta_{\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$
T2	$+4\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(5\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$
T3	$-9g_2^2\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$	$+4\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(5\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$	$-9g_2^2\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$

Terms 2 of 3

Column	Representative term structure
T1	$-3g_Y^2\lambda_{S_1^\dagger S_2 S_2^\dagger^2}$
T2	$+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger^2} + 6\lambda_{S_1 S_1^\dagger^2 S_2^\dagger})$
T3	$+\lambda_{12}^{\text{Adj}}\left(\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6} + \frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_Y^2\lambda_{S_1^\dagger S_2 S_2^\dagger^2}$	$+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger^2} + 6\lambda_{S_1 S_1^\dagger^2 S_2^\dagger})$	$+\lambda_{12}^{\text{Adj}}\left(\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6} + \frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)$

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$
T3	$+12\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$+12\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\lambda_{S_2}$

Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis C_{ijab} . Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$$\beta_{C_{1133}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1133}}{d\mu}=\beta_{C_{1133}}^{(1)}$$

Column	Representative term structure
T 1	$\frac{\sqrt{3}lambdaT_3\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T 1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}lambdaT_3\overline{y_1}y_2}{3MF}$

$$\beta_{C_{1134}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{1134}}{d\mu}=\beta_{C_{1134}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$

$$\beta_{C_{1144}}^{(1)}$$

$$16\pi^2\,_{\mu}\frac{dC_{1144}}{d\mu}=\beta_{C_{1144}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1155}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1155}}{d\mu}=\beta_{C_{1155}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta_{C_{1156}}^{(1)}$

$_{16\pi^2}^2\mu\frac{dC_{1156}}{d\mu}=\beta_{C_{1156}}^{(1)}$

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3ig_2^2\overline{y_1}^2}{4MF}$	$-\frac{iye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta_{C_{11511}}^{(1)}$

$16\pi^2\mu\frac{dC_{11511}}{d\mu}=\beta_{C_{11511}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}\overline{\lambda}12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}\overline{\lambda}12Inv2\overline{y_1y_2}}{3MF}$	$\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$

$$\beta_{C_{11512}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11512}}{d\mu}=\beta_{C_{11512}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1}y_2}{4MF}$	$-\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1}y_2}{3MF}$	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{4MF}$

$\beta_{C_{1166}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1166}}{d\mu}=\beta_{C_{1166}}^{(1)}$

Column	Representative term structure
T1	$\frac{\textit{lambdaS1Inv1}\overline{y_1}^2}{MF}$
T2	$\frac{g_2^2\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{\textit{lambdaS1Inv1}\overline{y_1}^2}{2MF}$	$\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$

$$\beta_{C_{11611}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11611}}{d\mu}=\beta_{C_{11611}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ie\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ie\overline{y_1y_2y\overline{e}}}{4MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$

$\beta_{C_{11612}}^{(1)}$

$16\pi^2\mu\frac{dC_{11612}}{d\mu}=\beta_{C_{11612}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}\overline{\lambda}12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}\overline{\lambda}12Inv2\overline{y_1y_2}}{3MF}$	$\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$

$$\beta_{C_{111111}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{111111}}{d\mu}=\beta_{C_{111111}}^{(1)}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$

$\beta_{C_{111112}}^{(1)}$

$_{16\pi^2}{}_{\mu}\frac{dC_{111112}}{d\mu}=\beta_{C_{111112}}^{(1)}$

Column	Representative term structure
T1	$\frac{{ig_2^2\overline{y_2}}^2}{MF}$
T2	$\frac{ilambdaS2Inv1\overline{y_2}}{MF}^2$
T3	$\frac{iye\overline{y_2}}{MF}^2\overline{ye}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_2}}{4MF}^2$	$-\frac{ilambdaS2Inv1\overline{y_2}}{2MF}^2$	$\frac{iye\overline{y_2}}{4MF}^2\overline{ye}$

$\beta_{C_{111212}}^{(1)}$

$_{16\pi^2}^{\mu}\frac{dC_{111212}}{d\mu}=\beta_{C_{111212}}^{(1)}$

Column	Representative term structure
T1	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{g_2^2\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$

$$\beta_{C_{1213}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{1213}}{d\mu}=\beta_{C_{1213}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\sqrt{3}lambda T_3 \overline{y_1 y_2}}{3MF}$

$$\beta_{C_{1214}}^{(1)}$$

$$16\pi^2\,_{\mu}\frac{dC_{1214}}{d\mu}=\beta_{C_{1214}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$

$$\beta_{C_{1223}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{1223}}{d\mu}=\beta_{C_{1223}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$

$$\beta_{C_{1224}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{1224}}{d\mu}=\beta_{C_{1224}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{\sqrt{3}lambda T_3 \overline{y_1 y_2}}{3MF}$

$$\beta_{C_{1257}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{1257}}{d\mu}=\beta_{C_{1257}}^{(1)}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1}^2}{8MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{ye\overline{y_1}^2\overline{ye}}{8MF}$

$\beta_{C_{1258}}^{(1)}$

$$16\pi^2\,\mu\,\frac{dC_{1258}}{d\mu}=\beta_{C_{1258}}^{(1)}$$

Column	Representative term structure
T1	$\frac{i\lambda S1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iy e\overline{y_1}^2\overline{y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{i\lambda S1Inv1\overline{y_1}^2}{4MF}$	$\frac{3ig_2^2\overline{y_1}^2}{8MF}$	$-\frac{iy e\overline{y_1}^2\overline{y e}}{8MF}$

$\beta_{C_{1259}}^{(1)}$

$16\pi^2\mu\frac{dC_{1259}}{d\mu}=\beta_{C_{1259}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{ye\overline{y_1y_2y\overline{e}}}{8MF}$	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{6MF}$

$\beta_{C_{12510}}^{(1)}$

$16\pi^2\mu\frac{dC_{12510}}{d\mu}=\beta_{C_{12510}}^{(1)}$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ie\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{ie\overline{y_1y_2y\overline{e}}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$

$\beta_{C_{1267}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1267}}{d\mu}=\beta_{C_{1267}}^{(1)}$

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{3ig_2^2\overline{y_1}^2}{8MF}$	$-\frac{iye\overline{y_1}^2\overline{ye}}{8MF}$

$\beta_{C_{1268}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1268}}{d\mu}=\beta_{C_{1268}}^{(1)}$

Column	Representative term structure
T1	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{g_2^2\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{3g_2^2\overline{y_1}^2}{8MF}$	$-\frac{ye\overline{y_1}^2\overline{ye}}{8MF}$

$\beta_{C_{1269}}^{(1)}$

$16\pi^2\mu\frac{dC_{1269}}{d\mu}=\beta_{C_{1269}}^{(1)}$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1}y_2}{6MF}$	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{8MF}$

$\beta_{C_{12610}}^{(1)}$

$16\pi^2\mu\frac{dC_{12610}}{d\mu}=\beta_{C_{12610}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{ye\overline{y_1y_2y\overline{e}}}{8MF}$	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{6MF}$

$\beta_{C_{12711}}^{(1)}$

$16\pi^2\mu\frac{dC_{12711}}{d\mu}=\beta_{C_{12711}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}\overline{\lambda}12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}\overline{\lambda}12Inv2\overline{y_1y_2}}{6MF}$	$\frac{ye\overline{y_1y_2y\bar{e}}}{8MF}$

$$\beta_{C_{12712}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12712}}{d\mu}=\beta_{C_{12712}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1}y_2}{6MF}$	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{8MF}$

$\beta_{C_{12811}}^{(1)}$

$16\pi^2\mu\frac{dC_{12811}}{d\mu}=\beta_{C_{12811}}^{(1)}$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ie\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{ie\overline{y_1y_2y\overline{e}}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$

$\beta_{C_{12812}}^{(1)}$

$$16\pi^2\mu\frac{dC_{12812}}{d\mu}=\beta_{C_{12812}}^{(1)}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}\overline{\lambda}12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}\overline{\lambda}12Inv2\overline{y_1y_2}}{6MF}$	$\frac{ye\overline{y_1y_2y\bar{e}}}{8MF}$

$\beta_{C_{12911}}^{(1)}$

$16\pi^2\mu\frac{dC_{12911}}{d\mu}=\beta_{C_{12911}}^{(1)}$

Column	Representative term structure
T1	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{g_2^2\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{3g_2^2\overline{y_2}^2}{8MF}$	$-\frac{ye\overline{y_2}^2\overline{ye}}{8MF}$

$\beta_{C_{12912}}^{(1)}$

$16\pi^2\mu\frac{dC_{12912}}{d\mu}=\beta_{C_{12912}}^{(1)}$

Column	Representative term structure
T1	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T3	$\frac{iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{3ig_2^2\overline{y_2}^2}{8MF}$	$-\frac{iye\overline{y_2}^2\overline{ye}}{8MF}$

$\beta_{C_{121011}}^{(1)}$

$_{16\pi^2}^2\,_{\mu}\frac{dC_{121011}}{d\mu}=\beta_{C_{121011}}^{(1)}$

Column	Representative term structure
T1	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T3	$\frac{iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{3ig_2^2\overline{y_2}^2}{8MF}$	$-\frac{iye\overline{y_2}^2\overline{ye}}{8MF}$

$\beta_{C_{121012}}^{(1)}$

$_{16\pi^2}^2\mu\frac{dC_{121012}}{d\mu}=\beta_{C_{121012}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_2}^2}{8MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{ye\overline{y_2}^2\overline{ye}}{8MF}$

$$\beta_{C_{2211}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{2211}}{d\mu}=\beta_{C_{2211}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$

$$\beta_{C_{2212}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{2212}}{d\mu}=\beta_{C_{2212}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$

$$\beta_{C_{2222}}^{(1)}$$

$$16\pi^2\,_{\mu}\frac{dC_{2222}}{d\mu}=\beta_{C_{2222}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{2277}}^{(1)}$

$16\pi^2\,_{\mu}\frac{dC_{2277}}{d\mu}=\beta_{C_{2277}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta_{C_{2278}}^{(1)}$

$16\pi^2\mu\frac{dC_{2278}}{d\mu}=\beta_{C_{2278}}^{(1)}$

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3ig_2^2\overline{y_1}^2}{4MF}$	$-\frac{iye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta_{C_{2279}}^{(1)}$

$16\pi^2\,_{\mu}\frac{dC_{2279}}{d\mu}=\beta_{C_{2279}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ye\overline{y_1y_2y\overline{e}}}{4MF}$	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{3MF}$

$\beta_{C22710}^{(1)}$

$16\pi^2\mu\frac{dC_{22710}}{d\mu}=\beta_{C_{22710}}^{(1)}$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ie\overline{y_1y_2y\bar{e}}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ie\overline{y_1y_2y\bar{e}}}{4MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$

$\beta_{C_{2288}}^{(1)}$

$16\pi^2\mu\frac{dC_{2288}}{d\mu}=\beta_{C_{2288}}^{(1)}$

Column	Representative term structure
T1	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{g_2^2\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta_{C_{2289}}^{(1)}$

$16\pi^2\mu\frac{dC_{2289}}{d\mu}=\beta_{C_{2289}}^{(1)}$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1}y_2}{4MF}$	$-\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1}y_2}{3MF}$	$\frac{iy_e\overline{y_1}y_2\overline{y_e}}{4MF}$

$\beta_{C_{22810}}^{(1)}$

$16\pi^2\mu\frac{dC_{22810}}{d\mu}=\beta_{C_{22810}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ye\overline{y_1y_2y\overline{e}}}{4MF}$	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{3MF}$

$\beta_{C_{2299}}^{(1)}$

$16\pi^2\mu\frac{dC_{2299}}{d\mu}=\beta_{C_{2299}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$

$\beta_{C_{22910}}^{(1)}$

$16\pi^2\mu\frac{dC_{22910}}{d\mu}=\beta_{C_{22910}}^{(1)}$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{ilambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{iye\overline{y_2}^2\overline{ye}}{4MF}$

$\beta_{C_{221010}}^{(1)}$

$_{16\pi^2}^2\mu\frac{dC_{221010}}{d\mu}=\beta_{C_{221010}}^{(1)}$

Column	Representative term structure
T1	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{g_2^2\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$